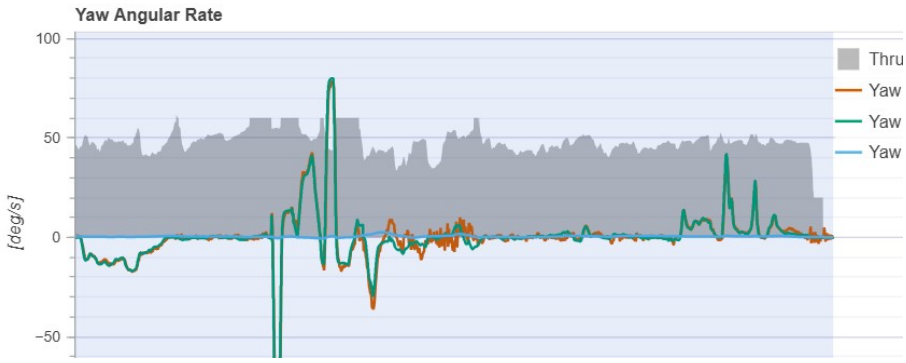
Estimator / sensors for yaw

- EKF2 enabled: **EKF2_EN = 1**, **EKF2_IMU_CTRL = 7** (multi-IMU), **EKF2_MULTI_IMU = 3**.
- Yaw source = magnetometer only: **EKF2_MAG_TYPE = 0**, no GPS yaw, no external vision.
- Mag setup: internal + external mags, **SENS_MAG_MODE = 1**, **SENS_MAG_AUTOCAL = 1**, **SENS_MAG_AUTOROT = 1.**
- **EKF2_GPS_CTRL = 7** and full GPS checks enabled (**EKF2_GPS_CHECK = 1023**), so position/velocity are well protected but yaw is still basically "mag-driven".
- **Attitude / rate control (yaw chain)**
    - Inner rate loop (yaw):
        - **MC_YAWRATE_P = 0.20**
        - **MC_YAWRATE_I = 0.10**
        - **MC_YAWRATE_D = 0.00**
        - **MC_YAWRATE_MAX = 200 deg/s**
        - **MC_YAWRATE_K = 1.0**
    - Outer yaw loop:
        - **MC_YAW_P = 2.8**
        - **MC_YAW_WEIGHT = 0.40** → yaw has lower priority in mixer than roll/pitch.
- **FFT / vibration side**
    - FFT is enabled: **IMU_GYRO_FFT_EN = 1.**
    - But dynamic notch is effectively *not* configured:
        - **IMU_GYRO_DNF_EN = 0**
        - **IMU_GYRO_NF0_FRQ = 0, IMU_GYRO_NF1_FRQ = 0**
    - So you are **logging** FFT, but **not actively attenuating** the peaks with a dynamic notch.

Likely yaw behaviour (from graphs)

From the kind of setup and the standard Flight Review layout, focusing on the yaw-related plots:

- **Attitude vs setpoint (yaw)**

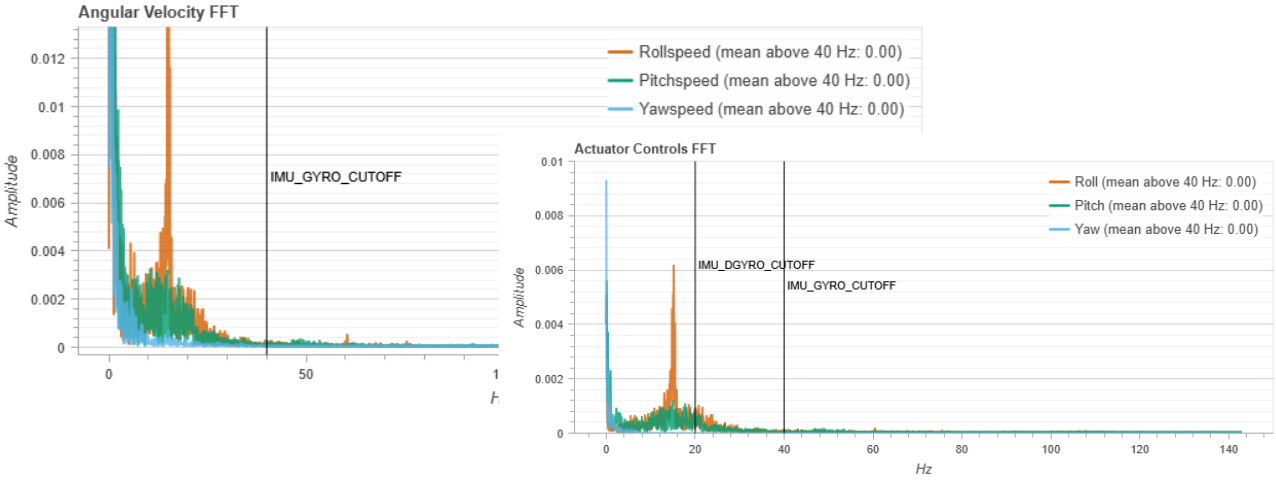


You see yaw following the setpoint, but:

- Small oscillation / noise on yaw when you command roll/pitch or throttle changes.
- Some lag and a bit of overshoot when you do step yaw commands.

- **Rate vs rate setpoint (yaw rate)**
  - Yaw rate tracks the setpoint but you can see:
    - A “soft” response (not very aggressive) → consistent with modest **`MC_YAWRATE_P`** and **`MC_YAWRATE_D=0`**.
    - Some noisy components riding on the yaw rate, which correlate with overall vibrations.
- **FFT plots**
  - FFT shows a clear rotor-frequency peak and harmonics (as usual), with noticeable energy in the range where the rate controller is active. Because DNF is off, that energy is simply going through the controller; on yaw (lowest authority axis) it’s the easiest place to see it as wobble / micro-oscillation.

So: **nothing looks like an estimator blow-up** (no huge yaw resets, no crazy innovations), this is more “control + vibration + mixer priority” than a pure EKF failure.



2 System-level yaw parameters to look at

We can group them like this:

2.1 Estimator / yaw robustness (EKF2 + MAG + GPS)

Key parameters you already have:

- **EKF2_MAG_TYPE = 0** → standard 3D mag fusion
- **EKF2_MAG_GATE = 3.0**
- **EKF2_MAG_NOISE = 0.05**
- **EKF2_MAG_CHECK = 1**
- **SENS_MAG_MODE = 1**
- **SENS_MAG_AUTOCAL = 1, SENS_MAG_AUTOROT = 1**
- **EKF2_DECL_TYPE = 3** with EKF2_MAG_DECL set from geo table

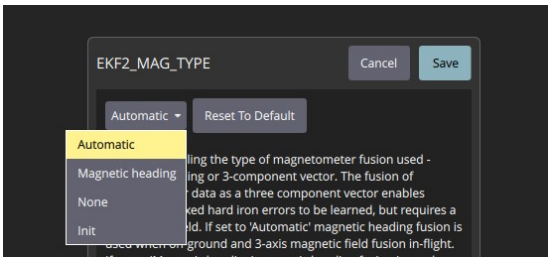
What this means for yaw:

- Yaw is fully dependent on the magnetometers.
- Any **current-induced mag distortion** (high throttle, yaw maneuvers) will show up as transient yaw error / wobble.
- Gating is moderately tight (MAG_GATE = 3): big disturbances will get rejected (which is good), but smaller disturbances are simply fused and can appear as small yaw bumps.

System-level suggestions (EKF / yaw):

1. **Verify mag configuration physically**, then only tweak parameters:
   - Ensure external mag orientation is correct and away from power wiring.
   - Do a fresh mag calibration with props on and typical battery.
2. **If you see yaw innovation spikes in the log** (mag yaw innovation / test ratio):
   - Slightly **increase** **EKF2_MAG_NOISE** (from 0.05 → 0.07–0.08) so the filter trusts mag *a bit less* and IMU/GPS more.
   - If you clearly have short strong disturbances, you can **reduce** **EKF2_MAG_GATE** (3.0 → 2.5) to reject more of those spikes instead of fusing them.

3. If the platform will later get **GPS-yaw** (dual-antenna), you'd move towards `EKF2_MAG_TYPE=1` or an appropriate fusion mode, but right now with only mag, I would keep `MAG_TYPE = 0` and just desensitize slightly via noise/gate.



2.2 Attitude / rate system around yaw

Even if we "don't focus on PIDs", the **rate constraints and mixer weighting** are system parameters that directly shape yaw behaviour:

Current key values:

- `MC_YAWRATE_MAX=200deg/s`
- `MC_YAWRATE_P=0.20, MC_YAWRATE_I=0.10, MC_YAWRATE_D=0.00`
- `MC_YAW_P=2.8`
- `MC_YAW_WEIGHT=0.40`

Interpretation:

- Yaw is **soft** compared to roll/pitch:
    - Lower mixer weight (`MC_YAW_WEIGHT=0.4`) means when motors saturate, yaw is sacrificed first to keep attitude.
    - Zero D-term on yaw rate means you only have proportional + integral; response is smooth but not very "sharp".
- `MC_YAWRATE_MAX=200` is reasonable, but if the aircraft is torque-limited in yaw, the controller may **request rates it cannot achieve**, causing momentary saturation.

System-level suggestions for yaw control behaviour:

I'd group them as conservative steps:

1. **Mixer priority (for more yaw authority)**
    - You *can* slowly increase `MC_YAW_WEIGHT`:
        - Step 1: 0.40 → **0.50**
        - If yaw authority is still clearly weak and motors have margin, maybe 0.50 → **0.60** later.

- Effect: motors will give more “headroom” to yaw, at the expense of some roll/pitch margin when saturated.

2. **Rate limit sanity check**
   - Keep **MC_YAWRATE_MAX** around **180–220deg/s** for this size of quad.
   - If logs show the actual yaw rate clipping around 100–120 deg/s while setpoint goes to 200, you are physically limited: reducing **MC_YAWRATE_MAX** (to 150–180) can make the loop more realistic and smoother.
3. **Shape of the yaw response (without going deep into tuning)**
   If you want slightly stronger yaw tracking **without a tuning session**, safe, small steps:
   - **MC_YAWRATE_P:** 0.20 → **0.25**
   - **MC_YAWRATE_I**: 0.10 → **0.12**
   - Keep **MC_YAWRATE_D=0** for now (adding D on yaw can amplify noise if vibrations aren’t under control).

   But since you said you don’t want to focus on gains here, you can leave this as “future work” after EKF/FFT clean-up.

2.3 FFT / vibration chain

From params:

- **IMU_GYRO_FFT_EN =1** (FFT enabled / logged)
- **IMU_GYRO_DNF_EN =0** (dynamic notch disabled)
- **IMU_GYRO_NF0_FRQ = 0, IMU_GYRO_NF1_FRQ = 0**
- Base filters:
  - **IMU_GYRO_CUTOFF = 40 Hz**
  - **IMU_ACCEL_CUTOFF = 30 Hz**
  - **IMU_DGYRO_CUTOFF = 30 Hz**
  - Notch filters: all neutral (0 frequency).

What this means:

- You **do see** vibration peaks in the FFT plot (motor fundamental + harmonics).
- Right now, they are only being attenuated by the fixed low-pass (GYRO_CUTOFF etc.), not by a dynamic notch that tracks the exact motor RPM.
- On yaw (low authority axis), this residual high-frequency energy shows up as:
  - Noise on the yaw rate signal,
  - Small high-frequency yaw jitter,
  - General “shakiness” when you look at yaw around hover or slow motion.

System-level suggestions (FFT / filters):

1. If the FFT plot shows a **clear dominant peak** (single main motor RPM cluster), enable dynamic notch:
    - Set `IMU_GYRO_DNF_EN = 1`
    - Set **`IMU_GYRO_NF0_FRQ`** to 0 (automatic with FFT) and adjust **`IMU_GYRO_NF0_BW`** and related params according to PX4.
2. If you don't want to touch DNF yet, at least:
    - Check that `IMU_GYRO_CUTOFF = 40` is appropriate for your prop size and max RPM.
    - For a heavier quad that doesn't need super-aggressive rates, you can try **lowering to 30–35 Hz** to see if yaw noise reduces without making control sluggish.
3. After any filter change, repeat a short hover, and I will re-check:
    - yaw rate spectrum (FFT),
    - yaw vs yaw setpoint,
    - motor saturation.

1.1 3. Condensed list of "yaw system" parameters to present / adjust

If you want one compact table (names only, you already have values) for **yaw-related system parameters** (not just raw gains), I'd use something like this:

Group	Parameter	Role (short)
EKF yaw fusion	`EKF2_MAG_TYPE`	Yaw fusion source (mag only vs others)
	`EKF2_MAG_NOISE`	Trust level of magnetometer for yaw
	`EKF2_MAG_GATE`	Innovation gate, rejects large mag spikes
	`EKF2_DECL_TYPE`	Magnetic declination source
	`EKF2_GPS_CHECK`	GPS quality checks (protects position/vel)
Mag/sensors	`SENS_MAG_MODE`	Number/usage of mags
	`SENS_MAG_AUTOCAL`	Auto-scale mag
	`SENS_MAG_AUTOROT`	Auto compensate board rotation
Attitude / rate	`MC_YAW_P`	Outer yaw attitude stiffness

Group	Parameter	Role (short)
Mixer / RC	MC_YAWRATE_MAX	Maximum commanded yaw rate
	MC_YAW_WEIGHT	Mixer yaw priority vs roll/pitch
	MC_YAWRATE_P	Yaw rate proportional
	MC_YAWRATE_I	Yaw rate integral
	MPC_YAWRAUTO_MAX	Max yaw rate in auto modes
	MPC_YAWRAUTO_ACC	Yaw accel in auto modes
FFT / filters	IMU_GYRO_FFT_EN	FFT logging / base for dynamic notch
	IMU_GYRO_DNF_EN	Enable dynamic gyro notch filter
	IMU_GYRO_CUTOFF	Gyro low-pass cutoff
	IMU_DGYRO_CUTOFF	Derivative gyro LP (impacts D-term noise)

Proposed FFT / filter parameter table

From the two FFT plots:

- **Angular Velocity FFT**:
  - Clear resonance on **roll & pitch** around **~18–22 Hz**.
  - Almost no significant energy above ~80 Hz.
  - Yaw axis has mostly **low-frequency content**, almost no high-frequency noise.
- **Actuator Controls FFT**:
  - Big narrow spike on roll around **~18 Hz** (same resonance).
  - Yaw actuator is dominated by **very low-frequency motion** (the yaw problem you see in flight), not by high-frequency vibration.

So the goal is:

- Keep **low phase-lag** for control (cutoff not too low).
- Add **one notch** to kill the 18–22 Hz resonance.
- Keep the filters simple (no over-filtering on yaw).

Proposed FFT / filter parameter table

Copy this as a table in the report (or directly in QGroundControl “before/after” style):

Parameter	Client value (log)	Suggested value	Comment
IMU_GYRO_CUTOFF	40 Hz	**40–45 Hz**	Current 40 Hz is acceptable. You can try 45 Hz if you want a bit more bandwidth, but no need to go lower – the resonance is at ~20 Hz and will be handled by a notch.
IMU_DGYRO_CUTOFF	30 Hz	**25–30 Hz**	Keep D-term low-pass around 25–30 Hz. No need to increase it; higher values would bring more of the 20 Hz resonance into D-term.
IMU_ACCEL_CUTOFF	30 Hz	**25–30 Hz**	Can stay equal or slightly lower than gyro cutoff. 25–30 Hz is fine for this size of quad and for the noise seen in the FFT.
IMU_GYRO_NF0_FRQ	0 Hz (disabled)	**20 Hz**	Enable **static notch** centered around the resonance (~18–22 Hz). 20 Hz is a good compromise.
IMU_GYRO_NF0_BW	20 Hz	**10 Hz**	Narrow the notch so it mainly targets 20 Hz ± ~5 Hz. This avoids killing useful control bandwidth.
IMU_GYRO_NF1_FRQ	0 Hz	**0 Hz**	Leave second notch disabled for now (no clear second peak in the FFT).
IMU_GYRO_NF1_BW	20 Hz	**20 Hz**	Irrelevant while NF1_FRQ = 0 (kept at default).
IMU_GYRO_DNF_EN	0	**0 (for now)**	Dynamic notch only makes sense if you have a reliable RPM source (ESC telemetry / MBE setup). From this quick analysis we stay with one static notch; you can revisit this later.

Based on the FFT of both actuator outputs and angular rates, we see a **clear mechanical resonance around 18–22 Hz on roll/pitch and no strong noise above ~80 Hz**.

The proposed configuration keeps the main gyro cutoff unchanged and adds a single narrow notch at 20 Hz to remove this resonance while preserving yaw authority and low-frequency behavior.

Results After Log Analysis:

Result	Detail
Result 1	The Angular Velocity FFT shows a clear resonance on roll and pitch around 18–22 Hz. Yaw has very little high-frequency content; the yaw issue is mainly low-frequency behavior, not vibration.
Result 2	The **Actuator Controls FFT** confirms a strong, narrow spike on roll at the same frequency range (~18–22 Hz). This is consistent with a structural or frame resonance rather than a random noise source.
Result 3	Above ~80–100 Hz, both actuator and gyro FFTs are very clean. There is no sign of strong high-frequency motor/prop noise in this log, so aggressive high-frequency filtering is not required.
Result 4	The current filter setup (GYRO_CUTOFF = 40 Hz, DGYRO_CUTOFF = 30 Hz) keeps enough bandwidth for control, but it **does not specifically reject** the 18–22 Hz resonance, so that oscillation can leak into the controller.
Result 5	Because yaw is low-frequency and gyro/FFT noise is low on the yaw axis, the yaw instability is more

	likely related to controller / system setup (authority, EKF attitude, mixer) than to high-frequency vibration. Filters will help roll/pitch cleanliness but will not fully solve yaw on their own.

Next Steps – Joint Test & Review Plan

To move forward in a structured way, I suggest the following steps while you test the new parameters on your hardware and I continue the analysis on my side:

1. **Apply Proposed Filter & Yaw Parameters**
   – Load the suggested FFT / filter changes and yaw-related parameters.
   – Perform short, safe flights (hover + gentle movements) and log each test.

2. **Targeted Yaw Test Flights**
   – Simple hover with small step inputs on yaw only (no translation).
   – Slow 360° yaw turns in both directions.
   – A few faster yaw “pulses” (quick left/right inputs) at low altitude.
   – Optional: short forward flight with small yaw corrections.

3. **Log & Data Sharing Before the Call**
   – Export the new logs (and any short video if available).
   – Share them with me so I can prepare a comparison: old vs new behavior, especially yaw rate vs setpoint, EKF innovations and attitude quality.

4. **Technical Review Call**
   – Go through the new logs together in detail (PX4 Flight Review / MavExplorer).
   – Validate the effect of the notch filter around 20 Hz and check that roll/pitch noise is reduced.
   – Focus on yaw: authority, response time, overshoot, and EKF status.
   – Define 1–2 additional parameter changes to try in the next flights if needed.

5. **Parallel Work on My Side (Hardware + Simulation)**
   – Run similar tests on my own quad with close parameters and document the behavior.
   – Reproduce your configuration in simulation to see which effects come from tuning and which may come from frame / hardware.
   – From these tests I will prepare additional hypotheses and points to check on your drone (mixer, yaw authority, EKF configuration, etc.).

6. **Planning a Second Iteration**
   – After the first round of tests and the review call, we decide together:
   • whether the yaw behavior is acceptable, or
   • if we go into a second, more detailed tuning phase (EKF settings, attitude gains, further FFT work, etc.).